Electrolysis Question Bank

Practice Questions with Worked Solutions

8 problems • 3 topics

Faraday's Law • Selective Discharge • Anodising

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Full worked-answer edition

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1. Faraday's Law: Deposition and Gas Evolution

Data for calculations. Take the Faraday constant as $F = 96\,500\,\mathrm{C\,mol^{-1}}$ and, where required, the molar volume of a gas at room conditions as $24.0\,\mathrm{dm^3\,mol^{-1}}$.

Question 1 — Mass of Copper Deposited

A current of 2.00 A is passed through $\mathrm{CuSO_4(aq)}$ using copper electrodes for 20.0 minutes. Calculate the mass of copper deposited at the cathode.

$$\mathrm{Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)}$$
$$A_r(\mathrm{Cu}) = 63.5$$

Solution

Charge passed:

$$\begin{aligned} Q &= It \\ Q &= (2.00)(20.0 \times 60) \\ Q &= 2400\,\mathrm{C} \end{aligned}$$

Moles of electrons:

$$\begin{aligned} n(e^-) &= \frac{Q}{F} \\ &= \frac{2400}{96500} = 0.0249\,\mathrm{mol} \end{aligned}$$

From the half-equation:

$$\mathrm{Cu^{2+} + 2e^- \rightarrow Cu}$$

$$2\,\mathrm{mol}\ e^- : 1\,\mathrm{mol\ Cu} \qquad n(\mathrm{Cu}) = \frac{0.0249}{2} = 0.0124\,\mathrm{mol}$$

Mass of copper:

$$m = nM = (0.0124)(63.5)$$

Answer

$$\boxed{m(\mathrm{Cu}) = 0.790\,\mathrm{g}}$$

Question 2 — Time Needed to Deposit Silver

A student wants to deposit 2.16 g of silver from $\mathrm{AgNO_3(aq)}$ using a current of 1.50 A. Calculate the time required.

$$\mathrm{Ag^+(aq) + e^- \longrightarrow Ag(s)}$$
$$A_r(\mathrm{Ag}) = 108$$

Solution

Moles of silver deposited:

$$n(\mathrm{Ag}) = \frac{2.16}{108} = 0.0200\,\mathrm{mol}$$

From $\mathrm{Ag^+ + e^- \rightarrow Ag}$, one mole of electrons deposits one mole of silver, so $n(e^-) = 0.0200\,\mathrm{mol}$.

Charge required:

$$Q = nF = (0.0200)(96500) = 1930\,\mathrm{C}$$

Time required:

$$t = \frac{Q}{I} = \frac{1930}{1.50} = 1287\,\mathrm{s}$$

Answer

$$\boxed{t = 21.4\,\mathrm{min}}$$

Question 3 — Volume of Hydrogen Produced

A current of 3.00 A is passed through dilute $\mathrm{H_2SO_4(aq)}$ using platinum electrodes for 15.0 minutes.
Calculate the volume of hydrogen gas produced at the cathode.

$$\mathrm{2H^+(aq) + 2e^- \longrightarrow H_2(g)}$$

Solution

Charge passed:

$$Q = It = (3.00)(15.0 \times 60) = 2700\,\mathrm{C}$$

Moles of electrons:

$$n(e^-) = \frac{2700}{96500} = 0.0280\,\mathrm{mol}$$

From $\mathrm{2H^+ + 2e^- \rightarrow H_2}$, two moles of electrons produce one mole of hydrogen.

$$n(\mathrm{H_2}) = \frac{0.0280}{2} = 0.0140\,\mathrm{mol}$$

Volume of hydrogen:

$$V = n \times 24.0 = (0.0140)(24.0) = 0.336\,\mathrm{dm^3}$$

Answer

$$\boxed{V(\mathrm{H_2}) = 0.336\,\mathrm{dm^3} = 336\,\mathrm{cm^3}}$$

Question 4 — Unknown Elements

A current is passed through two cells connected in series.
The first cell contains $\mathrm{XSO_4(aq)}$, while the second cell contains $\mathrm{Y_2SO_4(aq)}$.
The relative atomic masses of X and Y are in the ratio 1 : 3.
What is the ratio of the mass of X liberated to that of Y?

Solution

Since the two cells are connected in series, the same quantity of charge (and the same number of moles of electrons) passes through both.

Let

$$A_r(X) : A_r(Y) = 1 : 3, \qquad A_r(X) = M, \quad A_r(Y) = 3M.$$

For $XSO_4(aq)$, sulfate is SO_4^{2-}, so X is X^{2+}:

$$X^{2+} + 2e^- \rightarrow X.$$

Thus two moles of electrons deposit one mole of X.

For $Y_2SO_4(aq)$, each Y must be Y^+:

$$Y^+ + e^- \rightarrow Y.$$

Thus one mole of electrons deposits one mole of Y. If one mole of electrons passes through both cells,

$$m(X) = \frac{1}{2}M, \qquad m(Y) = 1 \times 3M = 3M.$$

Hence

$$m(X) : m(Y) = \frac{1}{2}M : 3M = 1 : 6.$$

Answer

$$\boxed{m(X) : m(Y) = 1 : 6}$$

Question 5 — Two Cells Connected in Series

Two electrolytic cells are connected in series.
Cell 1 contains $CuSO_4(aq)$, and copper is deposited:

$$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$$

Cell 2 contains $AgNO_3(aq)$, and silver is deposited:

$$Ag^+(aq) + e^- \longrightarrow Ag(s)$$

A current is passed until 0.635 g of copper is deposited in Cell 1.
Calculate the mass of silver deposited in Cell 2.

$$A_r(Cu) = 63.5, \qquad A_r(Ag) = 108.$$

Solution

In series, the same quantity of charge passes through both cells.

Moles of copper deposited:

$$n(Cu) = \frac{0.635}{63.5} = 0.0100\,\text{mol}$$

Since $Cu^{2+} + 2e^- \rightarrow Cu$, the moles of electrons passed are

$$n(e^-) = 2(0.0100) = 0.0200\,\text{mol}.$$

For silver, $Ag^+ + e^- \rightarrow Ag$, so $n(Ag) = 0.0200\,\text{mol}$.

$$m = nM = (0.0200)(108).$$

Answer

$$\boxed{m(\mathrm{Ag}) = 2.16\,\mathrm{g}}$$

2. Selective Discharge and Electrode Potentials

Question 6 — Order of Metal Plating

A solution at 25 °C contains 1.0 M Cd^{2+}, 1.0 M Ag^{+}, 1.0 M Au^{3+}, and 1.0 M Ni^{2+} in the cathode compartment of an electrolytic cell.
Predict the order in which the metals will plate out as the voltage is gradually increased.

Solution

At the cathode, reduction occurs. As the voltage is gradually increased, the ion with the most positive reduction potential is reduced most readily and plates out first.

Reduction half-equation	E / V	Plating order
$Au^{3+} + 3e^- \rightarrow Au$	$+1.50$	1st
$Ag^{+} + e^- \rightarrow Ag$	$+0.80$	2nd
$Ni^{2+} + 2e^- \rightarrow Ni$	-0.23	3rd
$Cd^{2+} + 2e^- \rightarrow Cd$	-0.40	4th
$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$	-0.83	after these metals

Water has a less positive reduction potential than the listed metal ions, so it will not interfere with plating of these metals under the simplified assumptions of the question.

Answer

$$\boxed{\mathrm{Au} \rightarrow \mathrm{Ag} \rightarrow \mathrm{Ni} \rightarrow \mathrm{Cd}}$$

Question 7 — Selective Plating under Non-standard Conditions

A solution at 25 °C contains 1.0 M Cu^{2+} and 1.0×10^{-4} M Ag^{+}.
Which metal will plate out first as the voltage is gradually increased when this solution is electrolysed?

Solution

Compare the reduction potentials under the given conditions.

For copper:

$$Cu^{2+} + 2e^- \rightarrow Cu, \qquad E = E^\circ = +0.34\,V,$$

because $[Cu^{2+}] = 1.0\,M$.

For silver:

$$\begin{aligned} Ag^+ + e^- &\longrightarrow Ag \\ E &= E^\circ - \frac{0.0591}{n} \log \frac{1}{[Ag^+]} \\ &= 0.80 - 0.0591 \log \frac{1}{1.0 \times 10^{-4}} \\ &= 0.80 - 0.0591(4) = 0.564\,V. \end{aligned}$$

Since $0.564\,V > 0.34\,V$, Ag^+ is reduced more readily than Cu^{2+}.

Answer

$\boxed{Ag(s) \text{ plates out first.}}$

3. Anodising Aluminium

Question 8 — Anodising Aluminium

The process of anodising aluminium objects is one of the most common applications of electrolysis.

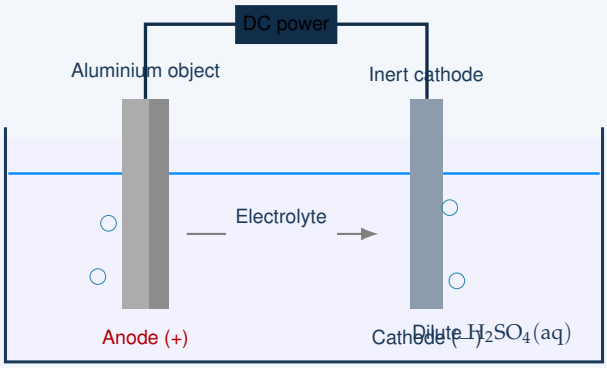


Figure 1: Schematic of anodising an aluminium object in dilute sulfuric acid.

The end result of this process is an increase in the thickness of the Al_2O_3 layer at the anode. However, consider the following standard electrode potential values:

$$Al^{3+}(aq) + 3e^- \longrightarrow Al(s) \qquad E = -1.66\,V$$
$$O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l) \qquad E = +1.23\,V$$

It would seem that the E values favour the preferential oxidation of aluminium at the anode. Instead, the process takes place via the oxidation of water first.
Offer an explanation. Finally, state the two reactions that occur at the anode to eventually form the final Al_2O_3 product.

Solution

Source note. The uploaded solution document states that its Question 8 model solution was added for completeness.

Although electrode potentials suggest that aluminium should be readily oxidised, the surface of aluminium is protected by a thin, strongly adherent Al_2O_3 layer. This passivating oxide layer creates a kinetic barrier and prevents rapid dissolution of aluminium as Al^{3+}.

Therefore, electrode potential values alone are insufficient: actual electrolysis also depends on kinetic factors, surface passivation, overpotential, and the nature of the electrode surface.

In the simplified mechanism presented in the supplied solution, water is oxidised at the anode to produce oxygen:

$$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-.$$

The oxygen formed at the aluminium surface reacts with aluminium to thicken the protective oxide layer:

$$4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s).$$

Answer

The two anode processes as stated in the supplied solutions are:

$$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$$
$$4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$$

The anodising process thickens the existing Al_2O_3 layer rather than simply dissolving aluminium into solution as Al^{3+}.

End of question bank